# Optimal Laser-Beam Engagement of Fast-Moving Inertial Fusion Energy Targets

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## 1 Introduction and Problem Statement

Inertial Fusion Energy (IFE) power plants rely on repetitively imploding millimeter-scale fuel capsules ("targets") by simultaneously illuminating them with an array of high-energy laser driver beams. In a reactor-scale facility, targets are injected at 400–500 m/s and must be engaged at chamber center with $\pm 20\,\mu$m accuracy [1, 3] at shot rates of 5–10 Hz, imposing severe constraints on both the tracking and beam-steering subsystems.

Two irreducible physical constraints bound the engagement window.

(i) *System latency* $\tau_\ell \in [1, 5]$ ms: the firing pipeline (signal processing, mirror actuation, pulse compression) introduces a fixed delay between the last measurement and beam firing.

(ii) *Laser steering rate*: each beamlet mirror satisfies $\|\dot{\boldsymbol{\theta}}_i\|_\infty \le \dot{\theta}_{\max}$ (rad/s), restricting repositioning speed. These two constraints together define the feasible command region and make this a nontrivial constrained stochastic optimization problem.

### 1.1 State and dynamics

Let the target state at time $t$ be

$$\mathbf{x}(t) = \left[\mathbf{r}^\top,\, \mathbf{v}^\top,\, \mathbf{a}^\top\right]^\top \in \mathbb{R}^9, \tag{1}$$

where $\mathbf{r}, \mathbf{v}, \mathbf{a} \in \mathbb{R}^3$ are position, velocity, and ballistic acceleration. Under a constant-acceleration model, the discrete-time dynamics are noised by a white gaussian noise $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{w}_k$$

where

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \Delta t\,\mathbf{I}_3 & \frac{\Delta t^2}{2}\mathbf{I}_3 \\ \mathbf{0} & \mathbf{I}_3 & \Delta t\,\mathbf{I}_3 \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 \end{bmatrix} \in \mathbb{R}^{9\times 9}.$$

### 1.2 Measurement model

At $m$ (= 3–8) discrete instants $t_1 < \cdots < t_m < t_f$, noisy observations $\mathbf{z}_k = \mathbf{H}\mathbf{x}_k + \boldsymbol{\nu}_k$, (with $\boldsymbol{\nu}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R})$), are acquired: transverse position via Poisson-spot shadow imaging ($\mathbf{H}$ selects the $xy$-rows of the state) and axial displacement via laser interferometry [2].

### 1.3 Optimization problem

As we expect very little laser corrections, the paraxial approximation holds. In such conditons, the beamlet $i$'s focal point is affine in $\mathbf{u}_i$ its pointing angles: $\mathbf{p}_i(\mathbf{u}_i) = \mathbf{L}_i\mathbf{u}_i + \mathbf{o}_i$, where $\mathbf{L}_i \in \mathbb{R}^{3\times 2}$ encodes path geometry (notably the distance between the mirror and the center of the reactor) and $\mathbf{o}_i$ is the nominal focus. Stacking $\mathbf{u} = [\mathbf{u}_1^\top, \ldots, \mathbf{u}_N^\top]^\top \in \mathbb{R}^{2N}$.

The probleme can be resumed as : *Find the beam-pointing command vector* $\mathbf{u} \in \mathbb{R}^{2N}$ *(azimuth/elevation angles for* $N$ *beamlets) that minimizes expected engagement error at the firing time* $t_f = t_{now} + \tau_\ell$*, subject to per-beamlet slew-rate limits.*

$$\min_{\mathbf{u}} \quad f(\mathbf{u}) = \mathbb{E}\left[\sum_{i=1}^{N} \left\|\mathbf{p}_i(\mathbf{u}_i) - \hat{\mathbf{r}}(t_f)\right\|^2 \,\middle|\, \mathbf{z}_{1:m}\right] \tag{2}$$

$$\text{s.t.} \quad \begin{cases} \|\mathbf{u}_i - \mathbf{u}_{i,\text{prev}}\|_\infty \le \dot{\theta}_{\max}\tau_\ell, \quad i = 1, \ldots, N, \\ \mathbf{u}_{\min} \le \mathbf{u} \le \mathbf{u}_{\max}. \end{cases}$$

where $\mathbf{p}_i(\mathbf{u})$ is the focal point of beamlet $i$ as a function of its pointing angles, $\hat{\mathbf{r}}(t_f)$ is the predicted target centroid at firing time, and the expectation is taken over the posterior distribution of $\mathbf{x}(t_f)$ given available measurements.

## 2 Proposed Approach

### 2.1 Forward Model: Kalman Filtering and Stochastic Reduction

The trajectory prediction model applies the Kalman filter (KF) to fuse the $m$ noisy observations. The filter propagates the posterior $p(\mathbf{x}_k|\mathbf{z}_{1:k}) = \mathcal{N}(\hat{\mathbf{x}}_{k|k}, \mathbf{P}_{k|k})$ via the standard predict–update cycle. At each measurement step $k$, the *Kalman gain*

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^\top\left(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^\top + \mathbf{R}\right)^{-1} \tag{3}$$

minimizes the posterior error covariance:

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\mathbf{H})\mathbf{P}_{k|k-1}$$

,

and the mean updates as

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}\hat{\mathbf{x}}_{k|k-1})$$

.

After $m$ update steps, the filter is propagated forward to $t_f$ to obtain $p(\mathbf{r}(t_f)|\mathbf{z}_{1:m}) = \mathcal{N}(\hat{\mathbf{r}}, \mathbf{P}_r)$, where $\hat{\mathbf{r}} \in \mathbb{R}^3$ and $\mathbf{P}_r \in \mathbb{R}^{3\times3}$ are the predicted position mean and covariance.

## 2.2 Optimization Strategy

### 2.2.1 Convexity and KKT sufficiency

The reduced objective $\tilde{f}(\mathbf{u}) = \sum_i \|\mathbf{L}_i\mathbf{u}_i - \tilde{\mathbf{r}}_i\|^2$ (with $\tilde{\mathbf{r}}_i = \hat{\mathbf{r}} - \mathbf{o}_i$) has block-diagonal Hessian

$$\nabla^2\tilde{f} = 2\begin{bmatrix} \mathbf{L}_1^\top\mathbf{L}_1 & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{L}_2^\top\mathbf{L}_2 & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \dots & \mathbf{L}_N^\top\mathbf{L}_N \end{bmatrix}. \tag{4}$$

Since each $\mathbf{L}_i \in \mathbb{R}^{3\times2}$ has full column rank (distinct beam geometries ensure that no two beamlets share a null direction), each block $\mathbf{L}_i^\top\mathbf{L}_i$ is symmetric positive definite, so the Hessian $\nabla^2\tilde{f}$ is symmetric positive definite, ensuring that $\tilde{f}$ is strictly convex. The feasible set $\mathcal{U}$ is a convex polytope. Choosing $\mathbf{u} = \mathbf{u}_{\text{prev}}$ as a Slater point (it strictly satisfies the slew-rate constraint with equality in the limit $\tau_\ell \to 0$, but is interior for any finite $\tau_\ell$), strong duality holds and the KKT conditions are necessary *and sufficient* for the global minimum.

### 2.2.2 Unconstrained solution and projection.

Absent constraints, each beamlet's optimal pointing has the closed form

$$\mathbf{u}_i^* = (\mathbf{L}_i^\top\mathbf{L}_i)^{-1}\mathbf{L}_i^\top(\hat{\mathbf{r}} - \mathbf{o}_i), \tag{5}$$

the least-squares pseudo-inverse. When $\mathbf{u}^* \in \mathcal{U}$, it is the global solution outright; otherwise, the constrained optimum is found via a log-barrier interior-point method. At each outer iteration, the barrier-augmented objective $\phi_\mu(\mathbf{u}) = \tilde{f}(\mathbf{u}) - \mu\sum_j \ln s_j(\mathbf{u})$ is minimized by Newton steps.

### 2.2.3 Stochastic extension.

If the target undergoes non-Gaussian perturbations (micro-vibrations, gas puff interactions), the Gaussian posterior is replaced by a sample-average approximation over $S$ Monte Carlo draws from an empirical noise model, following the stochastic gradient framework of [5]. Gradients are computed via automatic differentiation, and convergence of the SAA minimizer to the true optimum at rate $O(1/\sqrt{S})$ follows from standard stochastic programming theory.

# 3 Measures of Success

Success is assessed along two quantitative axes:

1. **Engagement accuracy.**
   RMS pointing error $\varepsilon_{\text{rms}} = \sqrt{N^{-1}\sum_i \|\mathbf{p}_i^* - \mathbf{r}(t_f)\|^2}$ over $10^3$ Monte Carlo trajectories. Target: $\varepsilon_{\text{rms}} \le 20\,\mu$m at $3\sigma$ [1], compared to polynomial-extrapolation and no-latency-correction baselines.
2. **Noise robustness.**
   Degradation of $\varepsilon_{\text{rms}}$ as $\text{tr}(\mathbf{R})$ is swept from nominal to $10\times$, and as $m$ drops from 8 to 5 (even to 2 measurements).

# 4 Prior Work and Literature

This project builds on the HAPL program's tabletop demonstration [2], which achieved $\pm20\,\mu$m accuracy at 4 m/s. Extending this to 400–500 m/s with latency-aware predictive control is the central contribution. Kalman filtering for state-vector trajectory estimation is classical [4]. Interior-point theory and Newton convergence bounds follow [6]; the SAA stochastic framework follows [5]. Physical parameters (4.5 mm capsule, 10 MJ laser) are drawn from Inertia Enterprises' public SPIE report [7].

# References


[1] R. Alexander et al., "Target injection and engagement for inertial fusion energy," *GA IFE Workshop*, 2022.

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